

HARSHAD PATEL

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Embedded Systems & IoT Architect | Remote Firmware Consultant

ESP32 | STM32 | Industrial Automation | 24+ Years Experience

PROFESSIONAL SUMMARY

Embedded Systems Architect with 24+ years of experience in firmware development, industrial automation, and IoT solutions. Expertise in ESP32, STM32, ARM Cortex platforms, and real-time embedded systems.

Delivered 25+ products including IoT gateways, SCADA systems, BMS, and remote monitoring solutions used in industrial and government applications.

Strong in remote debugging, embedded architecture, and protocol development (MODBUS, MQTT, CAN). Seeking remote opportunities as Embedded Firmware Developer, IoT Consultant, or Technical Architect.

CORE SKILLS

Embedded Systems

ARM Cortex (STM32, LPC), ESP32, RTOS (FreeRTOS)

Programming

Embedded C, C++, Firmware Architecture, Driver Development

Protocols

MODBUS (RTU/TCP), MQTT, CAN, TCP/IP, GSM/GPRS, GPS

Hardware

PCB Design, Industrial Interfaces, Power Systems

IoT & Remote Systems

Cloud Integration, Remote Monitoring, Data Logging

REMOTE WORK CAPABILITIES

- Remote debugging using logs, serial monitoring, OTA updates
 - MQTT cloud integration & remote device communication
 - ESP32 web server & dashboard development
 - Working with distributed teams
 - Documentation & async communication
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PROFESSIONAL EXPERIENCE

CEO & Embedded Systems Architect

Dazzle Technologies (Sep'09 – Present)

- Designed and delivered 25+ embedded and IoT products for industrial applications
 - Developed ESP32-based IoT gateway (MODBUS to MQTT cloud integration)
 - Implemented remote monitoring systems with real-time dashboards
 - Led hardware, firmware, and application teams (5–20 members)
 - Delivered SCADA, smart lighting, and automation solutions
 - Managed product lifecycle: concept → design → deployment
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Team Leader – Embedded Systems

CIMCON Software (Sep'06 – Aug'09)

- Designed PLC & RTU systems for SCADA applications
 - Developed embedded firmware for industrial automation
 - Led firmware team and testing activities
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Senior Project Engineer

Explora Infotech (Feb'04 – Aug'06)

- Developed smart card & access control systems
 - Designed electrical and networking systems for government projects
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R&D Engineer – Embedded Systems

INSTRUMENTS UNIVERSAL, VADODARA (Sep'02 – Jan'04)

- Designing, planning, developing and testing of microcontroller based products.
 - Developed embedded firmware for industrial automation.
 - Designed & developed Hardware, PCB & software.
 - Component selection, Part selection, Evaluating, Developing & testing of components for projects & products.
 - Site support & documentation.
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R&D Engineer – Embedded Systems

E CUBE INDIA SOLUTIONS LTD., VADODARA (Jul'01 – Aug'02)

- Designed & developed Mobile SMS, smart cards, Access control, Time attendance, and automatic vending machines based products.
 - Developed embedded firmware in C,C++ & Assembly.
 - Designed & developed Hardware, PCB & software.
 - Component selection, Part selection, Evaluating, Developing & testing of components for projects & products.
 - Site support & documentation.
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KEY PROJECTS (REMOTE / IoT FOCUS)

ESP32 IoT Gateway

- MODBUS RTU to MQTT cloud bridge
- Remote monitoring via web dashboard
- Secure data transmission

Battery Management System (BMS)

- STM32-based firmware
- CAN communication with ECU
- Real-time monitoring system

Remote Monitoring Unit (RMU)

- GSM/GPRS-based remote data acquisition
- Cloud alerts and logging

Vehicle Tracking System

- GPS + GSM real-time tracking
- Event-based reporting

TOOLS & TECHNOLOGIES

- Arduino IDE, PlatformIO, VS studio
- MQTT (Mosquitto, HiveMQ), IOT
- Basic Web: HTML, JavaScript
- Embedded Debugging Tools
- EasyEDA, Altium, Protel
- AI Tools – Chatgpt, geminiAI, Copilot, Claude

EDUCATION

B.E. Electronics & Communication Engineering
DDIT, Nadiad (2001) – Distinction